CONTACT TOPOLOGY WORLD MODELS: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

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ABSTRACT

We study contact-rich robot world models. The motivating bet is: Predict topology changes in contact graphs rather than only state trajectories.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For contact-rich robot world models, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2026a;b; 2025a;b; 2026c;d). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a, the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.520	0.480
observed_only_wam	hidden_mode_shift	0.175	0.825
augmentation_as_data_wam	nominal	0.635	0.365
augmentation_as_data_wam	hidden_mode_shift	0.240	0.760
scalar_uncertainty_wam	nominal	0.652	0.348
scalar_uncertainty_wam	hidden_mode_shift	0.403	0.597
proposed_branch_mechanism	nominal	0.787	0.213
proposed_branch_mechanism	hidden_mode_shift	0.527	0.473

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Contact Topology World Models argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

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